

Evidence-First Bengali Hallucination Detection: A Deterministic Selective Cascade

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Team Huntrix, IUT 12th ICT FEST Datathon, 2026

Abstract

We study selective inference for Bengali hallucination detection. Task-specific solvers, context grounding, and relation-typed evidence act first; a quantized open-weight model judges abstentions. Phase 1 public and private macro-F1 were 0.967 and 0.982. On 299 released labels, macro-F1 is 0.9933, with 225 deterministic rows classified without error. Three T4×2 runs reproduced the scored 2,516-row output. In the organizer-executed 5,000-row Phase 2 run, 4,793 rows reached the judge and inference took 3 hours 52 minutes (Team Huntrix, 2026). The results expose the trade-off between strict evidence admission, abstention, and runtime under source shift.

1 Introduction

The Bengali LLM Hallucination Detection Challenge asks whether a candidate response is fabricated, unsupported, or factually wrong (label=0), or faithful (label=1). A row contains a Bengali prompt, a response, and sometimes a supporting context. The released development set has 299 labels; Phase 1 contains 2,516 unlabeled rows. The held-out Phase 2 set contains approximately 5,000 rows from a wider range of sources (Ruhan et al., 2026; Competition Organizers, 2026).

This is not one homogeneous classification problem. The benchmark mixes context entailment, arithmetic, multiple-choice questions, morphology, idioms, legislation, history, literature, and civic information. It also inherits task diversity from Bengali hallucination research such as BenHalluEval (Adib et al., 2026). A generic LLM prompt must solve all of these relations with one evidence policy, and our early judges were both prompt-sensitive and expensive.

Our central decision was therefore to make the classifier selective. A deterministic component may label a row only when its scope, evidence

source, and semantic relation are explicit. Otherwise it abstains and sends the row to a bounded LLM judge. This separation gives exact tasks to exact methods and reserves model capacity for genuine ambiguity. The final system has three goals: high recall for the hallucinated class, protection against semantically related but irrelevant retrieval, and reproducible offline execution on Kaggle T4×2.

2 Method

Figure 1 shows the frozen v91 cascade. For input row x , the router returns either a label $d(x) \in \{0, 1\}$ with provenance or an abstention symbol $\perp$. The final prediction is

$$\hat{y}(x) = \begin{cases} d(x), & d(x) \neq \perp, \\ J(x), & d(x) = \perp, \end{cases}$$

where J is a fixed-order, grammar-constrained open-weight judge.

2.1 Normalization and Routing

The first stage normalizes Bengali and Latin digits, Unicode punctuation, spacing, answer aliases, and inline answer options. It also distinguishes absent context, evidential context, and instructional text. The last distinction matters because a grammar instruction can mention the requested concept without providing the answer.

Exact lineage checks match public benchmark or question-answer records only when normalized prompt, response, and applicable context fields agree. Reliability-weighted answer banks can resolve a row when the accepted answer relation is unambiguous. Approximate retrieval may nominate a candidate, but cannot overrule an exact source or an intrinsic context contradiction.

2.2 Structured and Context Solvers

Small solvers cover arithmetic, numeric ranges, Bengali compound number words, multiple-

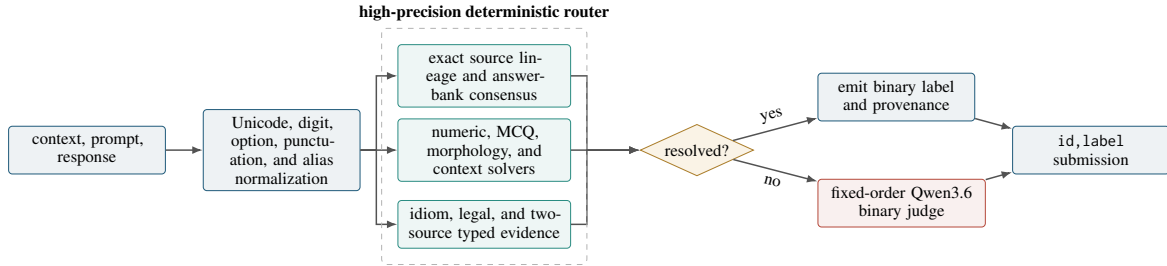


Figure 1: Frozen v91 inference path. Deterministic stages emit a label only for a relation-matched case; unresolved rows reach the open-weight judge.

Table 1: Evidence admission policy.

Evidence family	Requirement for emitting a label
Exact lineage	Prompt, response, and applicable context agree
Answer banks	Relation match and reliability-weighted consensus
Structured rule	Parsed values, entity, and predicate align
Context evidence	Same clause and prompt anchors; no contradiction
Idiom or law	Exact headword/section and requested sense/provision
Free-form fact	Typed relation and two independent sources
Fuzzy retrieval	Candidate generation only

choice serialization, morphology, compounds, and prefixes. Relation guards bind dates or numbers to the clause containing the requested entity and predicate. This prevents a valid year elsewhere in a passage from answering the wrong birth, death, publication, or establishment question.

For grounded rows, response claims are checked against the supplied passage. Exact answer spans are accepted only inside a semicolon-aware clause that also contains prompt anchors. Named entities, dates, negation, and alternatives are checked before external facts may contribute. Supplied context takes precedence over outside plausibility.

2.3 Evidence Admission and Conflict Resolution

Table 1 summarizes the admission policy. Idiom records distinguish figurative meaning from literal gloss, a known weakness in Bengali LLMs (Sakhawat et al., 2026). Legal records require the named act and section. A free-form fact requires two independent public sources and a typed relation. These guards were added after broad retrieval repeatedly found relevant passages that answered a neighboring question.

Every proposal records source family, relation, and confidence. Intrinsic context contradiction has highest priority. Exact authoritative lineage follows, then relation-aligned parsing and consensus evidence. If sources describe different relations, neither decides the row. If they claim the same relation but disagree, the system requires a stronger exact source or abstains. We accepted a larger judge queue when that was safer than allowing weak retrieval to override a trusted decision.

2.4 Selective Open-Weight Judge

The final Phase 1 router leaves 158 rows unresolved. They are sorted by row ID and evaluated in fixed batches. Context rows receive one entailment view. No-context rows receive up to three short views: factual verification, strict exam grading, and a zero-bias check. A faithful decision requires unanimous positive votes; evaluation stops after the first zero.

The judge is Qwen3.6-27B Q4_K_P, a 16.332 GiB GGUF checkpoint, served by pinned llama.cpp b9637 across two T4 GPUs (Qwen Team, 2026; Gerganov and contributors, 2026). Thinking is disabled, and a grammar restricts generation to one binary token. Seeds, row order, prompt order, and batch boundaries are fixed.

3 Experiments

3.1 Protocol

We used the 299 released labels for validation and error analysis, not as proof that a large fine-tune would generalize. We tracked macro-F1, per-class F1, accuracy, deterministic coverage, and context/no-context behavior. A candidate rule entered the system only when it expressed a reusable relation and passed the regression suite. The public leaderboard represented roughly 30% of Phase 1, so a rounded score change was treated as aggregate evidence rather than a row label.

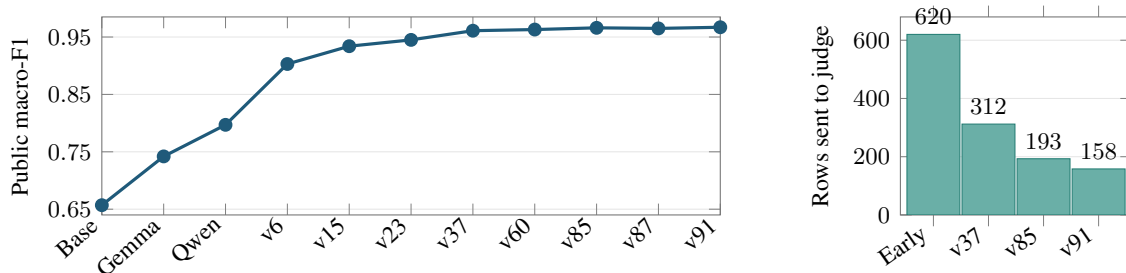


Figure 2: Left: public evaluation across selected versions. Right: the Phase 1 judge queue as routing rules matured. v6 marks the pivot from uniform judging to source- and task-conditioned decisions; v87 records a regression that was reverted.

Table 2: Representative scored iterations.

Version	Public F1	Main change
Baseline	0.657	character and embedding models
Gemma	0.742	12B QAT generic judge
Qwen	0.797	27B factual judge
v6	0.903	source and structured routing
v23	0.945	provenance repair
v37	0.961	source-lineage audit
v60	0.963	conservative fact corrections
v85	0.966	reproducible two-source anchor
v87	0.965	plausible edits; reverted
v91	0.967	typed idiom and relation guards

3.2 Iteration and Error Analysis

Figure 2 and Table 2 summarize the experiment path. Character and embedding models established a 0.657 baseline; Gemma and Qwen improved factual reasoning but remained prompt-sensitive. v6 shifted the hypothesis from model selection to decision routing. v37 tightened provenance, v87 exposed weakly typed facts and was reverted, and v91 added relation guards. The curve records pivots and reversions, not monotonic progress.

Several plausible approaches failed. Prompt voting reached 0.8512 locally but 0.783 publicly. TF-IDF reached 0.8261 out of fold but missed no-context cultural facts. QA retrieval often found the right entity and wrong relation; global substrings confused instructions with evidence. v87 made two plausible fact edits and regressed from 0.966 to 0.965, so we reverted it.

The decision log converted these failures into policy: retrieval became proposal-only, context rules became relation-aware, and facts required provenance plus a typed predicate. Admitted changes had to emit a trace and preserve 249 regression tests, including 225 deterministic controls.

Table 3: Validation and Phase 1 results.

Evaluation	Macro-F1	Other result
Released set (299)	0.993264	accuracy 0.993311
Deterministic (225)	1.000000	label-0 F1 1.000000
Phase 1 public	0.967	aggregate held-out split
Phase 1 private	0.982	aggregate final split

3.3 Results

Table 3 separates released-label validation from aggregate Phase 1 evaluation. The cascade makes two errors on 299 labels; both concern inconsistent controls that we did not patch row by row. The deterministic subset covers 225 rows without error. Public and private macro-F1 were 0.967 and 0.982. These aggregate scores neither identify row-level errors nor isolate the effect of each v91 rule.

4 Reproducibility and Phase 2

4.1 Frozen Inference and Determinism

The notebook reads the organizer’s literal path:

```
/kaggle/input/competitions/
bengali-hallucination/test set.csv
```

It does not search for alternate filenames, branch on row count, call an external API, or load a saved Phase 1 prediction vector. The organizer can replace the file mounted at that path without editing a variable.

Reproduction requires more than a seed. The notebook sorts unresolved IDs, fixes batch boundaries, pins both llama.cpp server commands, waits for health checks, and assigns each view to the same server. The CSV writer preserves incoming IDs and a fixed id, label column order. Three full runs produced byte-identical files matching scored submission 54836339.

Table 4: Frozen deployment measurements.

Property	Measurement
Hardware	Kaggle Tesla T4×2; offline
Model	Qwen3.6-27B Q4_K_P; 16.332 GiB
2,516-row runs	440.87, 527.03, 593.43 seconds
Phase 2 held-out run	13,921.91 seconds (3:52:01.91)
Phase 2 routing	207 deterministic; 4,793 judged
Phase 1 output	1,181 zeros; 1,335 ones
Regression suite	249 tests passing at freeze

4.2 External Resources

The evidence inventory combines public Bengali resources rather than treating any one corpus as universal truth. QA lineage came from TyDi QA, Suman Bengali QA, BUFFET, SQuAD-BN, BanglaRQA, IndicQA, Aya Indic, and public Bengali QA collections (Clark et al., 2020; Mondal, 2026; BUFFET Authors, 2023; CSE BUET NLP, 2026; Ekram, 2026; AI4Bharat, 2026; Cognitive Lab, 2026; Rasheduzzaman, 2026). Domain evidence included IndicXNLI, Bengali Wikipedia, textbooks, newspapers, Bangladesh legal corpora, history and literature QA, and public exam banks (Divyanshu, 2026; Wikimedia Foundation, 2026; Nishat, 2026; Nabil, 2026; Sakhawat, 2026b; Saha, 2026; Ramim, 2026; Lavlu, 2026; Badhon, 2026; Hishab, 2026; Wasi, 2026a,b). Bengali idiom and lexical records came from Bangla Bagdhara, BLUCK, BengaliDictionary, and a word-analogy collection (Sakhawat, 2026a; Abedin, 2026; Kamal, 2026; Mousumi, 2023). Wikidata supplied only corroborating structured facts (Wikidata Contributors, 2026). Sources with unclear redistribution terms were used for audit or retrieval, not copied into the public package.

4.3 Held-Out Domain Shift

The organizer preview names Common Crawl, Bengali Wikipedia, Banglapedia, government service pages, recent newspapers, legislation, NCTB textbooks, and Bengali literature (Competition Organizers, 2026). The completed 5,000-row held-out run confirmed the expected shift: deterministic coverage fell from 93.72% in Phase 1 to 4.14% in Phase 2, and 4,793 rows reached the judge (Team Huntrix, 2026). Table 5 states the fallback policy.

Unfamiliar sources bypass lineage and reach context reasoning or the judge; approximate matches are never forced. The measured Phase 2 run completed in 13,921.91 seconds, leaving 5 hours 7 minutes 58 seconds of headroom under

Table 5: Routing policy for announced Phase 2 sources.

Source family	Primary route; abstention trigger
Common	Context check or judge; unsupported outside claim
Crawl/news	Exact entity-relation match; alias ambiguity
Wikipedia/Banglapedia	Exact act and section; conflicting or absent provision
Government services	Clause, date, fee, and office checks; stale page
Legislation	Exact act and section; conflicting or absent provision
NCTB/literature	Structured solver or exact work relation; edition mismatch
Unknown	Skip lineage and invoke the fixed judge

the nine-hour limit.

No fine-tuned checkpoint is required. The package pins the public Qwen quantization, CUDA runtime, evidence dataset, and inference notebook. The router contains short, citation-backed facts from public sources, but no hidden labels or manually assigned row-label vector. Every output is recomputed from the input row and frozen assets.

5 Conclusion

The experiments indicate that performance depended more on matching the decision method to the task than on enlarging the model. Exact provenance and typed solvers handled cases with clear relations; the fixed judge handled the uncertain tail. The resulting pipeline reproduced its Phase 1 output and completed the 5,000-row Phase 2 run within the runtime limit. Its traces also expose which rule fired, which relation was checked, and where the system abstained.

Limitations

The private score does not guarantee equal accuracy on the broader Phase 2 source mix. Exact lineage cannot replace general reasoning on unfamiliar civic, legal, textbook, and news content, and the 299 released labels are too small for reliable per-domain estimates. The completed notebook proves runtime compliance but contains no Phase 2 ground truth or accuracy score. Evidence can age, conflict, or carry licensing constraints; high-stakes civic use still requires human escalation.

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